

Name: _____

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2009

EXTENSION 1 MATHEMATICS

Instructions:

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start each question on a new page

Total Marks

- Attempt Questions 1 – 7
- All questions are of equal value

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total

Question 1**Marks**

- a) Simplify $\frac{4^n}{4^{n+1}-4^n}$ 1
- b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{3x}$ 1
- c) The polynomial $P(x) = x^4 + ax^3 + 2x - 4$ has a remainder of -7 when divided by $x + 2$. Find the value of a . 2
- d) Find the coordinates of the point P which divide the interval from $A(-1,5)$ to $B(6,-4)$ externally in the ratio 3: 2. 2
- e) Find to the nearest degree, the acute angle between the lines $x - y = 2$ and $3x + y = 5$. 2
- f) Find $\int x\sqrt{1-x} \, dx$ using the substitution $u = 1 - x$ 2
- g) Solve for x : $\frac{2x-3}{x-2} \geq 1$ 2

Question 2 (Start a new page)

- a) Differentiate with respect to x
- (i) $y = \ln\left(\frac{2x-3}{3x+2}\right)$ 2
- (ii) $y = \tan^3(3x + 5)$ 2
- (iii) $y = \cos^{-1}(\sin x)$ 2
- b) Find
- (i) $\int \frac{dx}{3+4x^2}$ 2
- (ii) $\int \frac{2}{\sqrt{1-16x^2}} dx$ 2
- (iii) $\int \sin^2 \frac{x}{2} dx$ 2

Question 3**Marks**

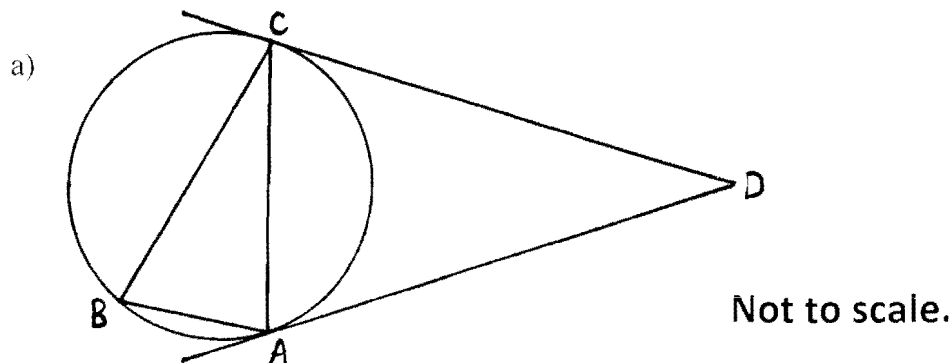
- a) Prove the identity $\frac{\cos x - \cos 2x}{\sin 2x + \sin x} = \operatorname{cosec} x - \cot x$ 3
- b) $P(x)$ is an odd polynomial of degree 3. It has $(x-2)$ as a factor and when it is divided by $(x-4)$, the remainder is 96. Find $P(x)$. 3
- c) Solve $\sqrt{3} \cos \theta - \sin \theta = -\sqrt{3}$ over the domain $0 \leq \theta \leq 2\pi$ 3
- d) Sketch $y = -2\sin^{-1} \frac{x}{3}$ showing the domain and range on your diagram. 3

Question 4

- a) Let T be the temperature inside a room at time t and let R be the constant outside air temperature. Newton's law of cooling states that the rate of change of the temperature T is proportional to $(T-R)$.
- (i) Show that the function $T = R + Ae^{-kt}$ (where A and k are constants) is a solution of the differential equation $\frac{dT}{dt} = -k(T - R)$ 1
- (ii) A metal baking dish is removed from an oven at 200°C . If the dish takes one minute to cool to 170°C , and the room temperature is 20°C , find the values of A and k , correct to 2 decimal places if necessary 2
- (iii) Find the time that it takes the dish to cool to 50°C . 1
- b) A particle is oscillating in simple harmonic motion about a fixed point. Its displacement $x\text{cm}$ at a time t seconds is given by $x = 2\cos 3t + 4$.
- (i) Explain why $2 \leq x \leq 6$ is the interval in which the particle moves. 1
- (ii) Write down the amplitude and centre of motion. 2
- (iii) Find $\ddot{x}$ as a function of t . 1
- (iv) Show that $\ddot{x} = -9(x - 4)$ 1
- (v) Show that $v^2 = -9x^2 + 72x - 108$ 2
- (vi) Find the greatest speed of the particle. 1

Question 5 (Start a new page)

Marks

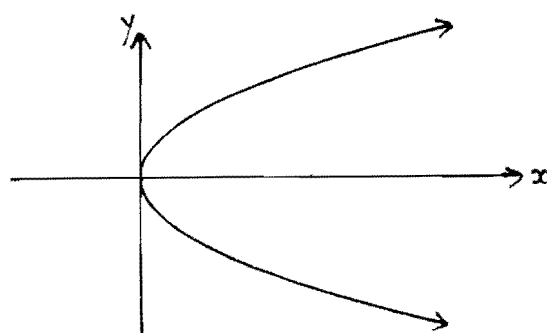


AD and CD are tangents to a circle. B is a point on the circle such that $\angle CBA$ and $\angle CDA$ are equal and are both double $\angle BCA$.

- (i) Copy the diagram into your answer booklet.
 - (ii) Let $\angle CDA = \alpha$ and derive $\angle CAD$ in terms of α (give reasons). 2
 - (iii) Prove that BC is a diameter of the circle (give reasons). 2
- b) The equation $x^3 - 2x^2 + 4x - 5 = 0$ has roots α, β, γ . Find the values of
- (i) $\alpha\beta\gamma$ 1
 - (ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1
 - (iii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ 2
- c) (i) Differentiate $x\cos^{-1}x - \sqrt{1-x^2}$ 2
- (ii) Hence evaluate $\int_0^1 \cos^{-1}x dx$ 2

Question 6 (Start a new page)

- a) Prove by Mathematical Induction for n a positive integer, that 4
- $$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$$
- b) (i) Find the equation of the normal at $P(at^2, 2at)$ on the parabola $y^2 = 4ax$. 2
- (ii) The normal intersects the x -axis at point Q . Find the coordinates of Q and hence find the coordinates of R where R is the midpoint of PQ . 2
- (iii) Hence find the Cartesian equation of the locus of R . 1



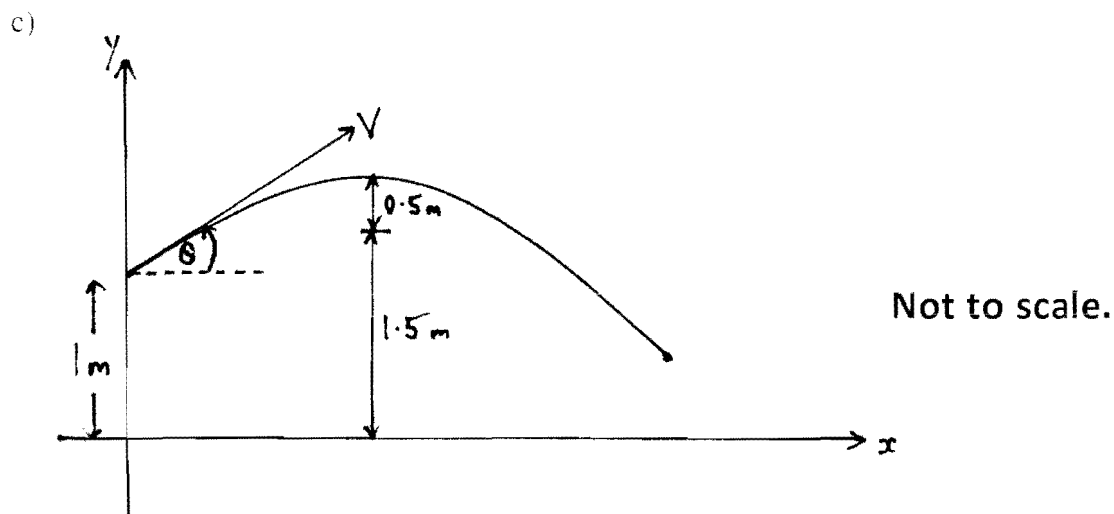
- c) Find $\int \frac{\cos x \sin x}{2 - \sin^2 x} dx$ using the substitution $u = \sin x$. 3

Question 7

Marks

- a) (i) Show that $\cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{1}{2}$ 1
- (ii) Hence prove that $\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$ 3

- b) When the polynomial $P(x)$ is divided by $x + 4$ the remainder is 5 and when $P(x)$ is divided by $(x - 1)$ the remainder is 9. Find the remainder when $P(x)$ is divided by $(x - 1)(x + 4)$. 3



A boy throws a ball and projects it with a speed of V m/s from a point 1m above the ground. The ball lands on top of a flowerpot in a neighbour's yard. The angle of projection is θ and indicated in the diagram. The equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -10$. It has been found that $y = Vt \sin \theta - 5t^2 + 1$.

- (i) Show that $x = Vt \cos \theta$ 1
- (ii) When the ball is at its maximum height above the ground, it passes directly above a 1.5 metre high fence and clears it by 0.5m. Show that $V = \frac{\sqrt{20}}{\sin \theta}$ 3
- (iii) Find the value of V given $\theta = \tan^{-1} \frac{9}{40}$, giving your answer in exact form. 1

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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$$a) \frac{4^n}{4^{n+1} - 4^n}$$

$$= \frac{4}{4^n(4-1)}$$

$$= \frac{1}{3} \quad \textcircled{1}$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 4x}{3x}$$

$$= \frac{4}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}$$

$$= \frac{4}{3} \quad \textcircled{1}$$

$$c) f(x) = x^4 + ax^3 + 2x - 4$$

$$-7 = (-2)^4 + a(-2)^3 - 4 - 4$$

$$1 = 16 - 8a \quad \textcircled{1}$$

$$8a = 15$$

$$a = \frac{15}{8} \quad \textcircled{1}$$

$$d) A(-1, 5) \quad B(6, -4)$$

$$m \cdot n = 3 \cdot -2 \text{ (external)}$$

$$3 \times 6 + -2 \times -1 \quad 3 \times -4 + -2 \times 5$$

$$x = \frac{3 \times -1}{3 \times -2} \quad \textcircled{1} \quad y = \frac{3 \times -2}{3 \times -2}$$

$$x = \frac{20}{1} \quad y = \frac{-22}{1}$$

$$\therefore P \text{ is } (20, -22) \quad \textcircled{1}$$

$$e) y = x - 2 \quad m_1 = 1$$

$$y = -3x + 5 \quad m_2 = -3$$

$$\tan \theta = \left| \frac{1 - (-3)}{1 + 1 \times -3} \right| \quad \textcircled{1}$$

$$= \left| \frac{4}{-2} \right|$$

$$\tan \theta = 2 \quad \textcircled{1}$$

$$\theta = 63^\circ \text{ (nearest } ^\circ \text{)}$$

$$f) \int 2\sqrt{1-x} \, dx \quad u = 1-x$$

$$du = -dx$$

$$\int (1-u)\sqrt{u} \cdot -du$$

$$= -\int u^{\frac{1}{2}} - u^{\frac{3}{2}} du \quad \textcircled{1}$$

$$= -\left(\frac{2}{3}u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}}\right)$$

$$= \frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + C \quad \textcircled{1}$$

$$g) \frac{2x-3}{x-2} \geq 1$$

Critical pts $x = 2$

$$\frac{2x-3}{x-2} = 1 \quad \textcircled{1}$$

$$2x-3 = x-2$$

$$x = 1$$

$$\frac{2x-3}{x-2} \geq 1 \quad \textcircled{1}$$

Question 2

$$a) w. y = \ln\left(\frac{2x-3}{3x+2}\right)$$

$$= \ln(2x-3) - \ln(3x+2) \quad \textcircled{1}$$

$$y' = \frac{2}{2x-3} - \frac{3}{3x+2} \quad \textcircled{1}$$

$$a) y = \tan^3(3x+5) \quad \textcircled{1}$$

$$y' = 3 \cdot 3 \tan^2(3x+5) \cdot \sec^2(3x+5)$$

$$= 9 \tan^2(3x+5) \sec^2(3x+5) \quad \textcircled{1}$$

①

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$$(iii) y = \cos^{-1}(\sin x)$$

$$y' = \frac{-1}{\sqrt{1-\sin^2 x}} \times \cos x \quad (1)$$

$$= \frac{-\cos x}{\sqrt{1-\sin^2 x}} \quad (1)$$

$$b. (i) \int \frac{dx}{3+4x^2}$$

$$\frac{1}{4} \int \frac{dx}{\left(\frac{\sqrt{3}}{2}\right)^2 + x^2} \quad (1)$$

$$\frac{1}{4} \times \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x}{\sqrt{3}}\right)$$

$$\frac{2\sqrt{3}}{4} \tan^{-1}\left(\frac{2x}{\sqrt{3}}\right) \text{ or } \frac{\sqrt{3}}{2} \tan^{-1}\left(\frac{2\sqrt{3}x}{3}\right) + C \quad (1)$$

$$(ii) \int \frac{2}{\sqrt{1-16x^2}} dx$$

$$\frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{1}{4}\right)^2 - x^2}} \quad (1)$$

$$\frac{1}{2} \sin^{-1} 4x + C \quad (1)$$

$$(iii) \int \sin^2 \frac{x}{2} dx$$

$$\int \frac{1-\cos x}{2} dx \quad (1)$$

$$\frac{x}{2} - \frac{\sin x}{2} + C \quad (1)$$

Question 3

$$a) \frac{\cos x - \cos 2x}{\sin 2x + \sin x} = \operatorname{cosec} x - \cot x$$

$$\frac{\cos x - (2\cos^2 x - 1)}{2\sin x \cos x + \sin x} \quad (1)$$

$$\frac{-2\cos^2 x + \cos x + 1}{\sin x (2\cos x + 1)} \quad (1)$$

$$\frac{(2\cos x + 1)(1 - \cos x)}{\sin x (2\cos x + 1)} \quad (1)$$

$$\operatorname{cosec} x - \cot x = \text{RHS}$$

$$c) \sqrt{3} \cos \theta - \sin \theta = -\sqrt{3}$$

$$\sqrt{3} \times \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = -\sqrt{3}$$

$$\sqrt{3}(1-t^2) - 2t = -\sqrt{3}(1+t^2)$$

$$2t = 2\sqrt{3}$$

$$\tan \frac{\theta}{2} = \sqrt{3}$$

$$\therefore \frac{\theta}{2} = \frac{\pi}{3}$$

$$\theta = \frac{2\pi}{3} \quad (1)$$

$$\text{Test } \theta = \pi$$

$$-\sqrt{3} - \sin \pi = -\sqrt{3} \checkmark$$

$$\therefore \text{Solutions are}$$

$$\theta = \pi, \frac{2\pi}{3} \quad (1)$$

b) $f(x)$ is odd. It must pass through $(0,0)$ and if $(x-2)$ is a factor, so is $(x+2)$ (1)

$$P(x) = ax(x-2)(x+2) \quad (1)$$

$$P(-4) = -4a(-6)(-2) = 96$$

$$a = -2$$

$$\therefore P(x) = -2x(x^2 - 4) \quad (1)$$

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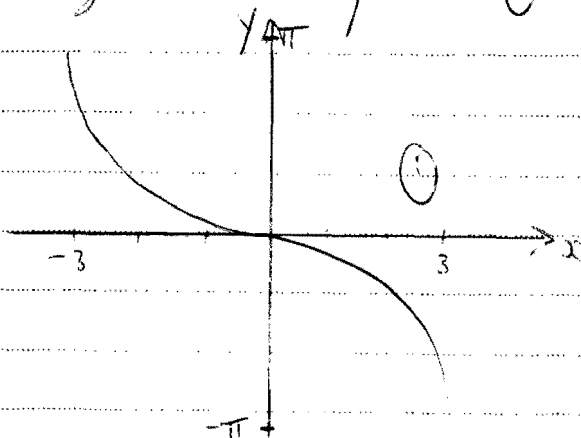
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$$d) y = -2 \sin^{-1}\left(\frac{x}{3}\right)$$

$$\text{Domain: } -1 \leq \frac{x}{3} \leq 1$$

$$-3 \leq x \leq 3 \quad (1)$$

$$\text{Range: } -\pi \leq y \leq \pi \quad (1)$$



Question 4

$$a) i) T = R + Ae^{kt}$$

$$\frac{dT}{dt} = Ae^{kt}$$

$$\frac{dT}{dt} = k \cdot Ae^{kt}$$

$$\frac{dT}{dt} = k(T - R) \quad (1)$$

$$ii) 70 = 20 + Ae^{k \cdot 1}$$

$$\text{also } 200 = 20 + Ae^0$$

$$180 = A \quad (1)$$

$$\therefore 170 = 20 + 180e^k$$

$$e^k = \frac{150}{180}$$

$$k = -0.18 \quad (1)$$

$$iii) T = 20 + 180e^{-0.18t}$$

$$50 = 20 + 180e^{-0.18t}$$

$$\frac{1}{6} = e^{-0.18t}$$

$$\ln \frac{1}{6} = -0.18t$$

$$t = \frac{\ln \frac{1}{6}}{-0.18}$$

$$t = 10.0 \text{ minutes} \quad (1)$$

$$b) \text{ c) Because } -1 \leq \cos 3t \leq 1$$

$$\therefore 2 \leq x \leq 4 \quad (1)$$

ii) Since motion is simple harmonic, centre of motion is halfway between 2 and 6
i.e. $x = 4 \quad (1)$
and amplitude is 2. (1)

$$iii) x = 2 \cos 3t + 4$$

$$\dot{x} = -2 \times 3 \sin 3t$$

$$\dot{x} = -18 \cos 3t \quad (1)$$

$$iv) x = -18 \cos 3t$$

$$= -9(2 \cos 3t + t - 4)$$

$$\dot{x} = -9(x - 4) \quad (1)$$

$$v) \dot{x} = \frac{d}{dt}\left(\frac{1}{2}v^2\right) = -9(x - 4)$$

$$\frac{1}{2}v^2 = -\frac{9x^2}{2} + 36x + C_1$$

$$v^2 = -9x^2 + 72x + C_2$$

$$\text{When } x = 2, v = 0 \quad (1)$$

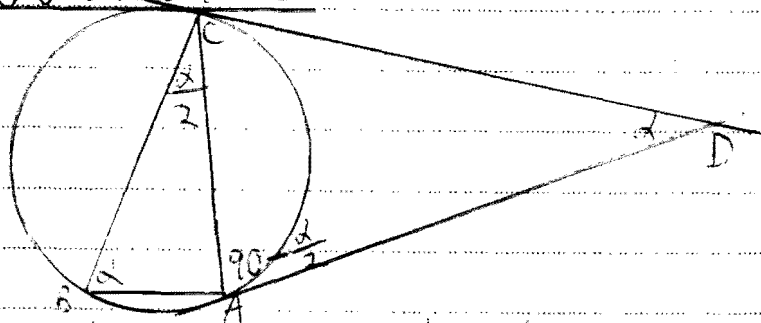
$$0 = -9 \times 2^2 + 72 \times 2 + C_2$$

$$C_2 = -108$$

$$\therefore v^2 = -9x^2 + 72x - 108 \quad (1)$$

vi) Greatest speed occurs at centre of motion $x = 4$.
 $v^2 = -9 \times 4^2 + 72 \times 4 - 108$
 $v_{\text{max}} = 6 \text{ cm/s} \quad (1)$

Question 5



(i) $\triangle CAD$ is isosceles
 $\angle CAD = 90 - \frac{x}{2}$ ①
 (equal base angles of an isosceles triangle)

(iii) $\angle ACD = 90 - \frac{x}{2}$ from (i)
 $\angle BCA = \frac{x}{2}$ (given)
 $\therefore \angle BCD = \frac{x}{2} + 90 - \frac{x}{2}$
 $= 90^\circ$ ①

has roots α, β, γ
 $\therefore \angle \beta\gamma = -\frac{d}{c}$
 $= 5$ ①

$\therefore BC$ is a diameter as
 angle between radius and
 pt. of contact of tangent
 CD is 90° ①

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$
 $= 4$ ①

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
 $= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$
 $= \frac{4}{5}$ ①

(iv) $\frac{d}{dx} (x \cos^{-1} x - (1-x^2)^{\frac{1}{2}})$
 $x \cdot \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x - \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot (-2x)$
 $= \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x + \frac{x}{\sqrt{1-x^2}}$
 $= \cos^{-1} x$ ①

(v) $\int_0^1 \cos^{-1} x \, dx$

$= \int_0^1 \frac{d}{dx} [x \cos^{-1} x - \sqrt{1-x^2}] \, dx$

$= [x \cos^{-1} x - \sqrt{1-x^2}]_0^1$ ①

$= 1 \times \cos^{-1} 1 - 0 - (0 - \sqrt{1})$

$= 1$ ①

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Question 6

a) Step 1 - Show result is true for $n=1$ ① for steps 1, 2
 $1 \times 2^0 = 1 + (1-1)2^1$
 $1 = 1$ ✓

Step 2 - Assume result is true for $n=k$, k an integer > 1 .

$$\text{i.e. } 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$$

Step 3 - Show result is true for $n=k+1$

$$\begin{aligned} \text{i.e. } & 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1) \times 2^k \\ & 1 + (k-1)2^k + (k+1) \times 2^k \quad \text{using Step 2} \quad \text{①} \\ & 1 + [k-1 + k+1]2^k \quad \text{①} \\ & 1 + [2k] \times 2^k \\ & 1 + k \times 2^{k+1} \quad \text{as required} \quad \text{①} \end{aligned}$$

Step 4

Since result is true for $n=1$, it must also be true for $n=1+1=2$, $n=2+1=3$ and hence for all positive integral values of n .

$$\begin{aligned} \text{b) i) } m_{\text{tangent}} &= \frac{dy}{dx} \\ &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ &= \frac{2a}{2at} \\ &= \frac{1}{t} \end{aligned}$$

$$\therefore m_{\text{normal}} = -t \quad \text{①}$$

$$y - 2at = -t(x - at^2)$$

$$+x + y = 2at + at^3 \quad \text{is ①}$$

the eq'n of normal

$$\text{ii) At Q, } y=0$$

$$\therefore +x = 2at + at^3$$

$$x = 2a + at^2$$

$$\therefore Q(2a + at^2, 0) \quad \text{①}$$

$P(at^2, 2at)$. R is midpoint of PQ

$$\text{i.e. } \left(\frac{2a + at^2 + at^2}{2}, \frac{0 + 2at}{2} \right)$$

$$R \text{ is } (a + at^2, at) \quad \text{①}$$

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$$\text{iii) } R(a + at^2, at)$$

$$y = at, \quad x = a(1+t^2)$$

$$t = \frac{y}{a}$$

$$\therefore x = a\left(1 + \frac{y^2}{a^2}\right)$$

$$ax = a^2 + y^2 \quad (1)$$

$$\text{c) } \int \frac{\cos x \sin x}{2 - \sin^2 x} dx$$

$$\text{Let } v = \sin x$$

$$dv = \cos x dx$$

$$\int \frac{v dv}{2 - v^2} \quad (1)$$

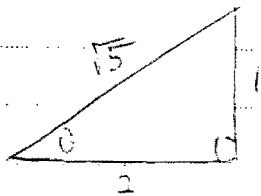
$$= \frac{1}{-2} \int \frac{-2v}{2 - v^2} dv$$

$$= -\frac{1}{2} \log_e (2 - v^2) + c \quad (1)$$

$$= -\frac{1}{2} \log_e (2 - \sin^2 x) + c \quad (1)$$

Question 7

a) i)



$$\text{Let } \theta = \cos^{-1} \frac{2}{\sqrt{5}}$$

From Δ above

$$\theta = \tan^{-1} \frac{1}{2}$$

$$\therefore \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{1}{2} \quad (1)$$

ii) Prove

$$\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$$

$$\tan^{-1} \frac{2}{3} + \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{7}{4}$$

from part i)

$$\text{Let } \alpha = \tan^{-1} \frac{2}{3}, \quad \beta = \tan^{-1} \frac{1}{2}$$

$$\therefore \tan \alpha = \frac{2}{3}, \quad \tan \beta = \frac{1}{2} \quad (1)$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{2}{3} + \frac{1}{2}}{1 - \frac{2}{3} \times \frac{1}{2}} \quad (1)$$

$$= \frac{\frac{7}{6}}{\frac{2}{3}}$$

$$\tan(\alpha + \beta) = \frac{7}{4} \quad (1)$$

$$\therefore \alpha + \beta = \tan^{-1} \frac{7}{4}$$

$$\therefore R.H.S = L.H.S$$

$$\text{b) Let } P(x) = (x-1)(x+4)Q(x) + ax + b \quad (1)$$

$$P(-4) = 0 - 4a + b = 5 \quad \text{Solve}$$

$$P(1) = 0 + a + b = 9 \quad \text{simultaneously}$$

$$5a = 4 \quad a = \frac{4}{5} \quad b = 8\frac{1}{5}$$

$$\therefore \text{Remainder is } \frac{4}{5}x + 8\frac{1}{5} \quad (1)$$

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c) (i) $\ddot{x} = 0$

$$\dot{x} = c$$

When $t=0$ $\dot{x} = V \cos \theta$

$$\therefore V \cos \theta = c$$

$$\dot{x} = V \cos \theta$$

$$x = Vt \cos \theta + c$$

when $t=0$ $x=0 \therefore c=0$

$$\therefore x = Vt \cos \theta \quad (1)$$

ii) At max. height,

$$y=2 \text{ when } \dot{y}=0$$

$$\dot{y} = V \sin \theta - 10t$$

$$0 = V \sin \theta - 10t$$

$$t = \frac{V \sin \theta}{10} \text{ when } y=2$$

$$y = Vt \sin \theta - 5t^2 + 1 \quad (1)$$

$$2 = V \sin \theta \cdot \frac{V \sin \theta}{10}$$

$$- 5 \times \frac{V^2 \sin^2 \theta}{100} + 1 \quad (1)$$

$$1 = \frac{V^2 \sin^2 \theta}{10} - \frac{5V^2 \sin^2 \theta}{100}$$

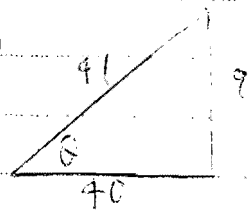
$$1 = \frac{10V^2 \sin^2 \theta - 5V^2 \sin^2 \theta}{100}$$

$$1 = \frac{V^2 \sin^2 \theta}{20}$$

$$V^2 = \frac{20}{\sin^2 \theta}$$

$$\therefore V = \frac{\sqrt{20}}{\sin \theta} \quad (1)$$

(iii)



$$V = \frac{\sqrt{20}}{\sin \theta}$$

$$= \frac{\sqrt{20}}{\frac{9}{41}}$$

$$= \frac{41\sqrt{20}}{9}$$

$$= \frac{82\sqrt{5}}{9} \text{ m/s} \quad (1)$$